

```
#include <iostream>
#include <string>
#include <algorithm>
```

```
using namespace std;
```

```
// We need to tell the compiler
about endFunction early so
startFunction can see it
int endFunction(string s, int i, int
j);
```

```
// This function checks if a piece
of the string is balanced
bool isBalance(string s, int start,
int end) {
    int frequency[26] = {0};
    int different_chars = 0;
```

```
// Count how many times each  
letter appears  
for (int i = start; i <= end; i++) {  
    int index = s[i] - 'a';
```

```
        if (frequency[index] == 0) {  
            different_chars =  
different_chars + 1;  
        }  
        frequency[index] =  
frequency[index] + 1;  
    }
```

// It must have exactly 2
different characters to be
balanced

```
    if (different_chars != 2) {  
        return false;  
    }
```

```
int value1 = 0;  
int value2 = 0;
```

```
// Check if the counts of the  
two characters are equal  
for (int i = 0; i < 26; i++) {  
    if (frequency[i] > 0) {  
        if (value1 == 0) {
```

```
        value1 = frequency[i];
    } else {
        value2 = frequency[i];
        break;
    }
}
}

return (value1 == value2);
}
```

```
// This function tries every
// possible starting point
int startFunction(string s, int i) {
    if (i == s.length()) {
        return 0;
    }
}
```

```
int option1 = endFunction(s, i,  
i);  
int option2 = startFunction(s, i  
+ 1);
```

```
    return max(option1, option2);  
}
```

// This function tries every possible ending point for a specific start

```
int endFunction(string s, int i, int j) {  
    if (j == s.length()) {  
        return 0;  
    }
```

```
    int currentLength = 0;
```

```
    if (isBalance(s, i, j)) {  
        currentLength = j - i + 1;  
    }
```

```
    return max(currentLength,  
endFunction(s, i, j + 1));  
}
```

```
// The main entry point for the  
logic
```

```
int longestBalanced(string s) {  
    return startFunction(s, 0);  
}
```

```
int main() {  
    string input;  
    cout << "Enter string: ";  
    cin >> input;  
  
    int result =  
    longestBalanced(input);  
    cout << "Longest Balanced  
Length: " << result << endl;  
  
    return 0;  
}
```